MILITARY SPECIFICATION

CAPACITORS, FIXED, SUPERMETALLIZED PLASTIC FILM DIELECTRIC, DIRECT CURRENT FOR LOW ENERGY, HIGH IMPEDANCE APPLICATIONS, HERMETICALLY SEALED IN METAL CASES, HIGH RELIABILITY, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for high reliability, supermetallized plastic film dielectric, fixed capacitors, hermetically sealed in metal cases. Capacitors covered by this specification have a failure rate (FR) level established in accordance with MIL-STD-690. This FR level is established at a 90-percent confidence level and maintained at a 10-percent producer's risk and, unless otherwise specified (see 3.1), is based on MIL-C-83421 life tests performed at maximum rated voltage at maximum rated temperature.

1.2 Classification. Capacitors covered by this specification are classified by the style, as specified (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder, Tin Alloy: Tin-Lead Alloy; and Lead Alloy.

MILITARY

MIL-C-39028 - Capacitors, Packaging of.
MIL-C-83421 - Capacitors, Fixed, Supermetallized Plastic Film Dielectric, (DC, AC, or DC and AC), Hermetically Sealed In Metal Cases, Established Reliability, General Specification For
MIL-C-83421/1 - Capacitors, Fixed, Supermetallized Plastic Film, Dielectric, DC and AC, Hermetically Sealed in Metal Cases, Established Reliability, Styles CRH01, CRH02, CRH03, CRH04, CRH05, CRH06, CRH07, CRH08, CRH09, and CRH00.
MIL-C-87217/1 - Capacitors, Fixed, Supermetallized Plastic Film Dielectric Direct Current, For Low Energy, High Impedance Applications, Hermetically Sealed in Metal Cases, High Reliability, Styles CHS01, CHS02, and CHS03.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Aeronautical Systems Division, ATTN: ENES, Wright Patterson AFB, OH 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-790 - Reliability Assurance Program for Electronic Parts Specifications.
MIL-STD-1276 - Leads for Electronic Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D92-78 - Flash and Fire points by Cleveland Open Cup.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Qualification. Capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3). In addition, the manufacturer shall obtain certification from the qualifying activity that the reliability assurance requirements of 4.1.1 have been met and are being maintained. Unless acquired from the manufacturer or his authorized distributor listed or approved for listing on the qualified products list, parts furnished under this specification shall not be considered as having met the requirements of this specification.

3.3 Reliability. Reliability of capacitors furnished under this specification shall be maintained in accordance with the procedures and requirements specified in MIL-STD-790 and MIL-STD-690 with details specified in 4.1.1 and 4.5.

3.3.1 Baseline documentation. Prior to granting of qualification approval, a baseline document in accordance with appendix A, shall be prepared by the manufacturer and approved by the qualifying activity.

3.4 Materials. The materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

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3.4.1 Impregnant. The impregnant, if used, shall be chemically inactive with respect to the capacitor element and the case. The impregnant, either in the state of original application or as a result of having aged, shall have no adverse effect on the performance of the capacitor. For liquid-impregnated capacitors, the same material shall be used for impregnating as is used for filling.

3.4.2 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion. Silver plating shall not be used in any external portions of these capacitors.

3.4.2.1 Dissimilar metals. Where dissimilar metals are used in intimate contact with each other, provision shall be made to provide protection against electrolysis and corrosion. The use of dissimilar metals in contact, which may tend toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), shall not be acceptable. However, metal plating or metal spraying of dissimilar metals to base metals to provide similar or suitable abutting surfaces will be permitted (for example, the spraying of copper on aluminum for soldering operations will be permitted). The use of dissimilar metals separated by insulating material will also be permitted.

3.4.2.2 Solder. Solder for electrical connections shall be in accordance with QQ-S-571.

3.5 Design and construction. Capacitors shall be of the design, construction, and physical dimensions as specified (see 3.1).

3.5.1 Capacitor element. The capacitor element shall consist of metallized plastic film. The capacitor dielectric may include uncoated dielectric tissues (plastic film).

3.5.1.1 Construction. All capacitors shall have extended electrode construction to minimize inductance.

3.5.2 Case. Each capacitor shall be enclosed in a hermetically-sealed metal case (magnetic or nonmagnetic, as specified (see 3.1)) which will prevent leakage of the impregnant and, in addition, will protect the capacitor element from moisture and mechanical damage under all the test conditions specified herein. The end seals shall be glass- or ceramic-to-metal seal.

3.5.2.1 Insulating sleeves. Shrink-fitted insulation shall be used for the sleeves, and shall lap over the ends of the capacitor body.

3.5.3 Axial-wire lead terminals. Axial-wire lead terminals shall be as specified (see 3.1).

3.6 Thermal shock. When tested as specified in 4.7.2, capacitors shall withstand the extremes of high and low temperatures without visible damage.

3.7 Burn-in (when specified, see 3.1). When tested as specified in 4.7.3, capacitors shall withstand the exposure to high temperature and voltage without visible damage.

3.8 High-impedance temperature and voltage ramp. When tested as specified in 4.7.4, capacitors shall withstand the extremes of high and low temperatures without visible damage. Any dc leakage current greater than 0.05 µA during ramp test shall be considered a failure.

3.9 Insulation resistance. When measured as specified in 4.7.5, the insulation resistance shall be not less than the values specified (see 3.1).

3.10 Capacitance. When measured as specified in 4.7.6, the capacitance shall be within the applicable tolerance specified (see 3.1).

3.11 Dissipation factor. When measured as specified in 4.7.7, the dissipation factor shall not exceed the value specified (see 3.1).

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3.12 Seal. When capacitors are tested as specified in 4.7.8, there shall be no repetitive bubbling, and for liquid-impregnated capacitors, there shall be no evidence of leakage.

3.13 Radiographic inspection. When the capacitors are radiographed in accordance with 4.7.9, there shall be no evidence of improperly made connections, substandard soldering, structural weakness, or attached solder particles or slivers (see appendix B).

3.14 High impedance dc life. When tested as specified in 4.7.10, capacitors shall meet the following requirements:

a. Insulation resistance (terminal to terminal): Shall meet the initial requirements (see 3.1).

b. Capacitance: Shall meet the initial requirements (see 3.1).

c. Dissipation factor: Shall meet the initial requirements (see 3.1).

d. Visual inspection: There shall be no corrosion, leakage of impregnant, or mechanical damage either during or after the test.

3.15 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

3.16 Marking (see 3.1). Marking of capacitors shall conform to method I of MIL-STD-1285, and shall include the part number, "JAN" marking, date code, lot symbol, manufacturer's source code, capacitance (in µF), capacitance tolerance, and rated voltage.

Example of marking

M87217/01-2107A - Part number (may be on one line if space permits).
.018 µF 5% 50 V - Capacitance, capacitance tolerance, and rated voltage.
JAN 8433A 12345 - "JAN" marking, date code, lot symbol and source code.
AAA - Serialized marking, each part, alpha or numeric (no limit on the number of spaces).

The manufacturer shall provide for lot traceability by date code and lot symbol. Manufacturing records shall include these same date codes and lot symbols.

3.16.1 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark "JAN".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or

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purchase order, the manufacturer may use his own or any other facilities suitable for the performance of the inspection requirements specified herein with approval by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790; "Program implementation" exceptions are as follows:

a. Under "Description of production processes and controls", the procedure for identification of each production lot shall include only "the manufacturer shall as a minimum be able to identify the time period during which the final production operation was performed on each item of product prior to final test. The date or lot code marked on each part shall be identified to a production lot."

b. "Traceability" of materials shall not apply.

Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4) via MIL-C-83421.

b. Verification of qualification (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions and methods.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202, except relative humidity shall not exceed 75 percent. Unless otherwise specified (see 3.1), accuracy of all test voltage measurements shall be within ±2.0 percent of the specified voltage.

4.3.2 Methods.

4.3.2.1 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at 25°C ±3°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.3 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the specified test voltage.

4.4 Qualification inspection. 1/ Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production. Parts shall be "S" failure rate level and selected from manufacturers qualified to M83421/1 "S" failure rate level.

4.4.1 Sample size. The number of capacitors to be subjected to qualification inspection shall be as specified in MIL-C-87217.

4.5 Verification of qualification. Every 6 months the manufacturer shall compile a summary of the results of quality conformance inspections in the form of a verification of qualification report, and forward it to the qualifying activity within 30 days of the end of the reporting period as the basis of continued qualification approval. In addition, the manufacturer shall immediately notify the

1/ Application for qualification tests shall be made in accordance with SD-6, "Provisions Governing Qualification" (see 6.3.1).

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qualifying activity whenever the data indicates that the manufacturer has failed to maintain his qualified product. Continuation shall be based on evidence that, over the 6-month period, the following has been met:

a. The manufacturer has not modified the design of the item.

b. The specification requirements for the item have not been amended so as to affect the character of the item.

c. Lot rejection for group A inspection does not exceed 5 percent or one lot, whichever is greater."

4.5.1 Records. Maintenance of life tests and FR level records shall be as specified in MIL-STD-690.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A.

4.6.1.1 Inspection lot. An inspection lot shall consist of capacitors of the same style, voltage rating, design and nominal capacitance rating produced in the same case size. Manufacture of all parts in the lot shall have been started, processed, assembled and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle.

4.6.1.2 Group A inspection. There shall be a maximum total of 5 percent defectives allowed (PDA) on the 100-percent tests of subgroups 2 and 3 for catastrophic failures. A catastrophic failure is defined as insulation resistance (IR) less than 100 megohms, capacitance value ±20 percent or greater of its nominal value, or dissipation factor (DF) of .25 percent or greater.

4.6.1.2.1 Sampling plan. Subgroup 1, 2, 3 and 4 tests shall be performed on each capacitor offered for inspection. Statistical sampling and inspection for subgroup 6 shall be in accordance with MIL-STD-105. The acceptable quality levels (AQL) shall be as specified in table I. Major and minor defects shall be as defined in MIL-STD-105 and as specified in table I.

4.6.2 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-39028.

4.7 Methods of inspection and test.

4.7.1 Visual and mechanical inspection. Capacitors shall be inspected to verify that the materials,, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.15 and 3.16). A 5X to 7X magnification limit shall apply.

4.7.2 Thermal shock (see 3.6). Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exception shall apply:

a. Test condition letter B, -55°C to + 100°C.

b. Measurements before and after cycling: Not applicable.

c. Number of cycles: As specified (see 3.1).

4.7.3 Burn-in (when specified, see 3.1) (see 3.7). Capacitors shall be exposed to the temperatures, voltages, and conditions as specified (see 3.1).

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TABLE I. Group A inspection

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of defectives permitted
Subgroup 1				
Thermal shock	3.6	4.7.2	(100% inspection)	
Subgroup 2				
Burn-in (when specified)	3.7	4.7.3		
High impedance temperature and voltage ramp test 2/	3.8	4.7.4	(100% inspection) 1/	
Subgroup 3				
Insulation resistance 3/	3.9	4.7.5		
Capacitance	3.10	4.7.6	(100% inspection) 1/	
Dissipation factor	3.11	4.7.7		
Subgroup 4				
Seal	3.12	4.7.8		
Radiographic inspection 4/	3.13	4.7.9	(100% inspection)	
Subgroup 5				
High impedance dc life	3.14	4.7.10		
Insulation resistance 5/	3.9	4.7.5		
Capacitance	3.10	4.7.6	20 6/	0
Dissipation factor	3.11	4.7.7		
			AQL (percent defective)	
Subgroup 6				
Visual inspection (external)	3.15	4.7.1	Major 1.0 (AQL)	Minor 4.0 (AQL)

1/ Percent defective allowable (PDA) = 5 percent.
2/ DC leakage failure if greater than .05 µA.
3/ At +25°C and +100°C with a 1-2 megohm series resistor, rated voltage.
4/ This test can be any place in sequence following subgroup 1.
5/ At +25°C rated voltage and initial limits.
6/ Parts may not be shipped.

4.7.4 High impedance temperature and voltage ramp (see figure 1).

a. Each part shall be connected with 1-10 megohms in series.

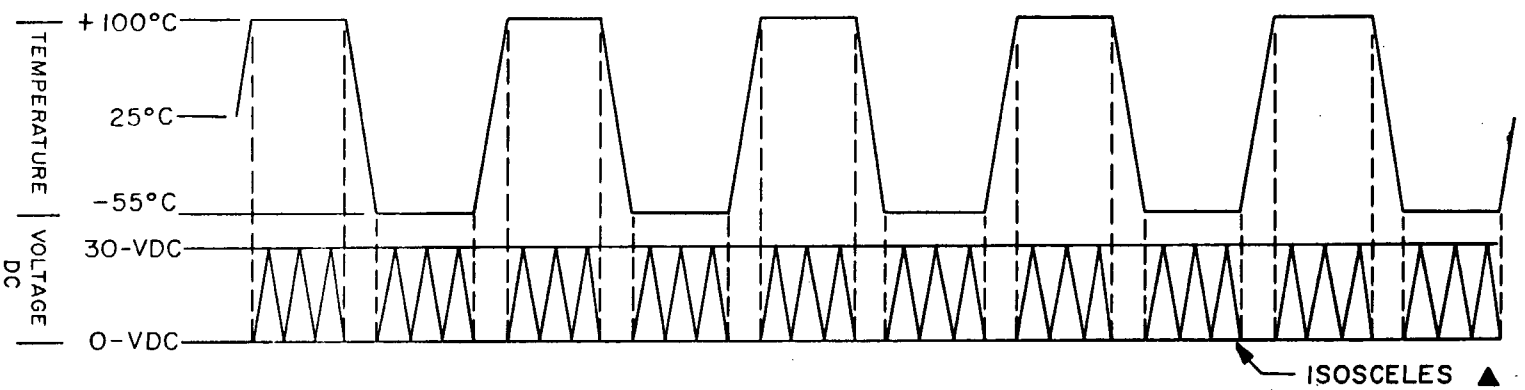
b. Parts shall be temperature ramped for 5 cycles from 25°C to 100°C, to -55°C, and to 25°C with an equally timed stay at the temperature extremes (100°C/-55°C).

c. Parts shall be voltage ramped during the temperature extremes only. A 0 to 30 volt (±10%) ramp in 30 minutes (±3 minutes) shall be applied to the RC circuit. After the capacitor reaches 30 volts charge, the capacitors will be linearly discharged within 30 ± 3 minutes. When the voltage reaches 200 millivolts, the voltage ramp to 30 volts is repeated (3 ramps during each exposure at the temperature extremes). For capacitance values above 10 microfarads, use a 60 ± 6 minutes ramp to 30 volts.

d. Parts shall be monitored for high leakage current continuously, or scanned at a minimum of once every 0.5 seconds, during the 5 temperature cycles. Noise shall be limited to 6 mV, peak.

e. A permanent record of current excursions in excess of .05 µA shall be maintained.

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STABILIZE CAPACITORS AT TEMPERATURE EXTREMES AND VOLTAGE RAMP 3 TIMES

(0 TO 30 V dc) ±10% AND (30 TO 0 V dc) ±10% WHILE MONITORING FOR LEAKAGE CURRENT

FIGURE 1. Voltage and temperature ramp test.

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4.7.5 Insulation resistance (see 3.9). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

a. Test potential: Rated voltage, unless otherwise specified (see 3.1).

b. Points of measurement:

Terminal to terminal: Between terminals at the applicable specified temperatures (see 3.1), and at 25°C ±3°C.

c. Charge time 5 minutes maximum; however, for capacitance values greater than 1.0 µF, an additional 1 minute per µF is permitted.

4.7.6 Capacitance (see 3.10). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. Unless otherwise specified (see 3.1), the following details shall apply:

a. Test frequency: 1,000 ±100 hertz (Hz).

b. Limit of accuracy: ±0.05 percent.

4.7.7 Dissipation factor (see 3.11). Unless otherwise specified (see 3.1), the dissipation factor shall be measured at 1,000 ±100 Hz (for capacitors having a nominal capacitance of 1 microfarad or less) or 100 ±10 Hz (for capacitors having a nominal capacitance greater than 1 microfarad).

4.7.8 Seal (see 3.12). Capacitors shall be tested in accordance with method 112 of MIL-STD-202. The following details shall apply:

a. Test condition letters: As specified (see 3.1).

b. Measurements after test: Not applicable.

4.7.8.1 Alternate test for liquid-impregnated capacitors only. Capacitors shall be placed with the terminals facing sideways (not upward) on a clean sheet of absorbent paper and exposed to a case temperature within +3°C, -0°C of the applicable maximum rated temperature (see 3.1) for a period of 4 hours minimum. After the test, capacitors and absorbent paper shall be visually inspected for evidence of leakage of impregnant.

4.7.9 Radiographic inspection (see 3.13). The capacitors shall be radiographed in accordance with appendix B. The following details shall apply:

a. Number of views: See appendix B, figure 4. Axial lead style capacitors: Two views perpendicular to the plane of the leads, with the second view made after rotation of the capacitors 90° around its own axis.

b. Inspection of films: The films shall be inspected on a back-lighted illuminator using a minimum of 10X magnification.

c. Defects: Capacitors containing any defects as specified in 3.13 and appendix B, figure 3 shall be rejected.

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d. Summary data of inspection results: A summary of the results of radiographic inspection (number of parts inspected, number accepted, number rejected) shall be included on the report of group A inspection furnished with each shipment of capacitors.

e. Retention of films: Radiograph films and test reports shall be retained by the manufacturer for a minimum period of 12 years.

4.7.10 High impedance dc life (see 3.14). The capacitors shall be tested 250 hours at 100°C with 1.5 ±0.5 volts dc and 1 through 10 megohms resistor in series with each capacitor so that the energy from source or across the capacitor is 500 microjoules, maximum.

a. Insulation resistance:

Terminal to terminal: Shall meet the initial requirements (see 3.1).

b. Capacitance: Shall meet the initial requirements (see 3.1).

c. Dissipation factor: Shall meet the initial requirements (see 3.1).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-39028 for government purchases.

6. NOTES

6.1 Intended Use. Capacitors covered by this specification are primarily intended for use in low-energy, high impedance applications such as integrators, RC timing circuits, precision filters and other applications where the ac component of voltage is not significant. Styles covered by this specification are available in tight tolerances and exhibit excellent capacitance stability over long life, as well as high insulation resistance and low loss factor. These capacitors can exhibit momentary breakdown in low energy applications. To insure clearing of breakdown, the application shall provide a minimum of 500 microjoules stored energy in the capacitor during operation. Applications for these capacitors shall be limited to circuits that can provide the minimum energy and that are insensitive to momentary breakdown/clearing actions.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet, and the complete part number and lead material if other than that specified (see 3.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the qualified products list is the Department of the Air Force; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC), Dayton, Ohio 45444. Application for qualification tests shall be made in accordance with SD-6, "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of SD-6, "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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PROCEDURES FOR THE PRODUCT AND QUALITY AUDIT

A10. SCOPE

A10.1 Scope. The appendix provides procedures to the Product and Quality Audit team for the auditing of manufacturers supplying to this specification for certification of manufacturing facilities. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

A30. INSTRUCTIONS

A30.1 Baseline document. The manufacturer shall complete the baseline document and submit the required information to the qualifying activity prior to scheduling the Product and Quality Audit. It is not intended that the manufacturer submit proprietary information. Each manufacturer shall provide a detailed process flow chart which shall include, as a minimum:

a. All significant process steps and measurement and inspection points by descriptive name, document number, revision letter, and revision date.

b. All raw material and interim and finished part quality control points by descriptive name, document number, revision letter, revision date, and location in the processing sequence.

c. All raw material incoming inspection, storage, and processing steps by descriptive name, document number, revision letter, revision date, and location in the processing sequence.

d. The identification of all critical (significant to end product properties) manufacturing and inspection steps as determined by the manufacturer in conjunction with the audit team at the time of baseline document approval.

e. Even though these parts were taken from MIL-C-83421 the above as a minimum applies.

A30.1.1 Changes to baseline. The audit team shall compare the latest baseline document to the previous baseline document to assure all changes have been submitted to the qualifying activity and have been approved.

a. Minor changes. Any subsequent departure from the approved flow chart (i.e., additions or deletions) or changes to the revision status of those processes or inspections are allowed, provided they do not violate the changes prohibited by Section b (major changes). These minor changes shall be described in general terms to the audit team during subsequent audits. This action shall result in an update of the baseline document.

b. Major changes. No changes (i.e., substitutions, additions, deletions or modifications) shall be allowed in the following areas without prior review and approval of the qualifying activity. In these, and other areas deemed major by the audit team at the time of baseline approval, partial or total product requalification may be required as determined by the qualifying activity.

1. Dielectric formulation and method of tape preparation.

2. Electrode material and method of application.

3. Mandrel material, design, and surface finish.

4. Film winding, including machines, film tensions, film speeds.

5. End termination materials, means of application, and subsequent product cleaning.

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A30.2 Audits. An initial audit and subsequent periodic audits (every year or as deemed necessary) shall be performed by the qualifying activity. The audit shall include, but not be limited to, all of the items on the checklist (plus MIL-STD-790) and any GIDEP ALERTS issued against the manufacturer at the facility of interest. The results of any additional items that are assessed shall be added to the checklist for record purposes.

A30.2.1 Duties of the audit team.

a. The audit team shall review the manufacturer's facility and the proposed baseline documentation for adequacy, accuracy and completeness during the audit of the manufacturer. The proposed baseline documentation should be used as a baseline/reference during the course of the audit.

b. The audit team shall verify that the ramp scan test equipment is in place, functioning and calibrated. They shall also verify that test measuring procedures and equipment are suffiently accurate for the required test.

c. The audit team shall typically sample check twelve months of records for the items being certified. In addition, the location of the equipment and the thoroughness and adequacy of the operating procedures and instructions shall be checked. Whenever practical, operations utilizing critical equipment and procedures shall be witnessed.

d. The audit team shall insure that adequate documentation, trained personnel, equipment, and visual aids are immediately available to perform the inspections specified in figure 2.

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Check - Item	Insp.	Satisfactory
a. Verification of MIL-STD-790 requirements		
b. Flow chart		
1. Key processes identified with revision.		
2. Key inspection points identified with revision		
3. Quality control points identified with revision		
4. Key raw material control documents identified with revision.		
5. Critical steps identified.		
6. Nonproprietary documents reviewed.		
7. Test and measurement points identified with revision.		
c. Baseline: Shall describe, as a minimum, the following:		
1. Capacitor design by part number indicating the following with the necessary tolerance defined:		
Dielectric film type. Dielectric thickness. Film width. Metallization thickness Metallization width. Metallization resistivity. Case length. Case diameter Additional material wound at start and stop.		
2. Documents listing suppliers qualified for all raw materials.		
d. Lot control plan (see 4.6.1.1)		
e. Traceability plan.		
f. Training plan for QC personel		

FIGURE 2. Certification check list.

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RADIOGRAPHIC INSPECTION

B10. SCOPE

B10.1 Scope. This appendix establishes the X-ray criteria for acceptance and rejection of hermetically sealed plastic dielectric capacitors. Typical examples of acceptable and unacceptable capacitors are shown on figure 3. This appendix is a mandatory part of the specification.

B10.2 Facilities. Radiographs of the capacitors may be made by the capacitor manufacturer or a suitable independent laboratory. In either case, facilities, equipment, personnel and techniques shall be in accordance with the requirements specified herein.

B10.3 Responsibility for inspection. The capacitor manufacturer shall perform the inspection of the radiograph films and acceptance/rejection of the capacitors.

B20. APPLICABLE DOCUMENTS

B20.1 Government document. The following standard forms a part of this appendix to the extent specified herein.

FED-STD-595 - Colors

B30. EQUIPMENT

B30.1 Dimensions. The dimensional values upon which these requirements are based were developed using the U S system. The metric equivalents were obtained by direct conversion based upon 1 inch = 25.4 mm. Where such equivalents do not correspond to a standard metric size, the closest standardized metric equivalent size may be used.

B30.2 Radiography. The X-ray equipment shall have sufficient voltage range to produce radiographs in accordance with this document. The equipment shall have a focal spot of 3.5 millimeters (mm) or less and shall maintain a sharply defined image at a focal film distance of 30 to 60 inches (76.20 to 152.40 centimeters).

B30.3 Exposure factors. The X-ray exposure factors shall be selected to achieve maximum image detail within the sensitivity requirements. The film shall be exposed in accordance with the following requirements:

a. X-ray voltage: Lowest voltage possible.

b. H and D film density: 1.0 to 2.5.

c. Milliampere and time settings: Adjusted, as necessary, to obtain satisfactory exposure.

B30.4 Film. The X-ray film shall be single emulsion and of a grade defined as very fine grain.

B30.4.1 Sensitivity. X-ray film and equipment shall be capable of detecting metallic particles with a major dimension of 0.001 inch (.03 mm) or greater.

B30.4.2 Exposure. Exposure factors such as KVP, current and time shall be compatible with the sensitivity requirements of B30.4.1.

B30.4.3 Film density. The X-ray equipment and processing techniques shall be capable of producing H and D film density of 1.0 to 2.5 in accordance with the American Standard Printing density type P-2.

B30.4.4 Film dimensions. Radiographic film shall not exceed 14 inches (35.56 centimeters) in width and 17 inches (43.18 centimeters) in length.

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B30.4.5 Processing. The exposed X-ray film shall be processed in such manner that the film shall be free of processing defects, i.e., fingerprints, chemical spots, blemishes, etc.

B30.4.6 Film identification. Each radiographic film shall be identified with the following information:

a. Part manufacturer's name.

b. Part number (as marked on part).

c. Part cross reference.

B40. REQUIREMENTS

B40.1 Views

B40.1.1 Tubular units. Two planes, normal to the major axis of the part, shall be taken. If deemed necessary by the testing laboratory or acquiring activity representative, additional views shall be required. Part orientation: See figure 4.

B40.2 General inspection. The inspection for all capacitors shall include, but not limited to, inspection for faulty lead connections, misalignment of internal parts, solder or weld defects and physical damage of electrical elements.

B40.2.1 Faulty lead connection. There shall be no evidence of improperly made lead connections.

B40.2.2 Misaligned parts. Criteria for rejection due to misalignment of capacitor section seals, eyelets or spacers shall be as specified in the detail inspections below.

B40.2.3 Solder defects. Criteria for rejection due to soldering defects shall be as specified in B40.3.2.

B40.2.4 Physical damage. There shall be no holes, cracks, splits or chips of electrical elements.

B40.3 Specific inspections. Inspections that are peculiar to plastic capacitors, in addition to the general inspections, are listed below. All capacitors failing to meet these requirements shall be rejected. Examples of typical defects, as well as acceptable devices, are shown.

B40.3.1 Extraneous material

B40.3.1.1 Any single foreign particle larger than .020 inch (.51 mm) in its longest dimensions, regardless of location, shall be cause to reject the part (see figure 3A). A string of particles less than .020 inch in diameter located between the section and case parallel to the axis of the section and which are continuous for more than 1/4 of the section length shall be cause to reject the part. Random and spray particles which are less than .020 inch (.51 mm) in diameter are typical of these capacitors and shall not be cause for rejection.

B40.3.1.2 Foreign particles, looped wire (unless characteristic of qualified design), metallization or excess solder from the end spray cap shall not extend into the capacitor element .050 inch or more when measured from the base of the end spray cap (see figure 3B).

B40.3.2 Spikes and excessive solder. Spikes of end spray or solder spikes other than those described above shall not extend through the winding insulation or protrude within .010 inch of capacitor case.

B40.3.3 Lead wire bends. Units having the internal lead wire bent/looped shall be rejected, unless bend/loop is characteristic of qualified design.

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B50. REPORTS AND RECORDS

B50.1 Reports of inspection. Unless otherwise specified by the contract, the testing activity shall furnish inspection reports signed by an authorized representative of the testing activity. The reports shall give the results of the radiographic inspection and shall list the purchase order number or equivalent identification, the part number, the number of parts inspected, the number of parts rejected and the date of the test. For each rejected part, the part number, the serial number and the cause for rejection of the part shall be listed. Two copies of the report shall be forwarded to the contracting activity.

B50.2 Records of inspection. A complete record of details of inspection shall be kept by the manufacturer or testing laboratory on a form acceptable to the preparing activity. The record shall list the voltage potentials and currents used in the radiographic process, the time of exposure, the distance of the source of radiation from the surface of the part, the distance of the film from the same surface, the approximate angle between the central beam of radiation and the film, the screens and filters used, the size of the focal spot, the time of development of the film and the serial number of the part under test. When an identical technique is used for a number of parts, a single record, tabulating all identical features, will suffice for all parts. Copies of the records shall be made available to the contracting activity when requested.

B50.3 Records of radiographs. Each radiograph shall carry a radiograph inspection serial number, or code letters, to identify the radiograph with the parts inspected shown in the radiograph. One copy of all radiographs shall accompany the shipment of parts when submitted to the contracting activity. In addition, one complete set of radiographs shall be kept by the manufacturer or testing laboratory for a period of 12 years.

B60. Personnel.

B60.1 Radiographer. Personnel engaged in radiographic processing shall be familiar with the requirements of this standard and with all other documentation controlling radiographic inspection of parts and materials. They shall be capable of producing radiographs which meet the requirements of all applicable documentation.

B60.2 Radiographic interpreters. Personnel engaged in the interpretation of radiographs shall be familiar with the requirements of this standard and with all applicable documentation controlling radiographic quality of parts and materials being inspected. They shall be capable of evaluating radiographs to determine conformance of parts and materials to the requirements of all applicable documentation.

B60.3 Vision. The minimum vision requirements for visual acuity of personnel inspecting film shall be as follows:

a. Distant vision shall equal 20/30 in at least one eye, either corrected or uncorrected.

b. Near vision shall be such that the individual can read Jeager type number 2 at a distance of 40.64 centimeters (16 inches), either corrected or uncorrected.

B60.4 Vision tests. Vision tests shall be performed by an oculist, optometrist or by other professionally recognized personnel. One year from the effective due date of qualification, and each year thereafter, qualified personnel shall be required to pass the vision test specified herein.

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The part shall be rejected if any one of the defects listed below is observed:

1. Length of eyelet to header less than 50% filled with solder. See Figure 3C for details.

2. Solder in header/case joint less than 50% of length of joint. See Figure 3C for details.

3. Loose particles larger than .020 inch in any dimension.

4. Particles larger than .020 inch in any dimension.

5. Shadowing due to plating solder or seal solder reflow is acceptable providing termination and lead wire is visible.

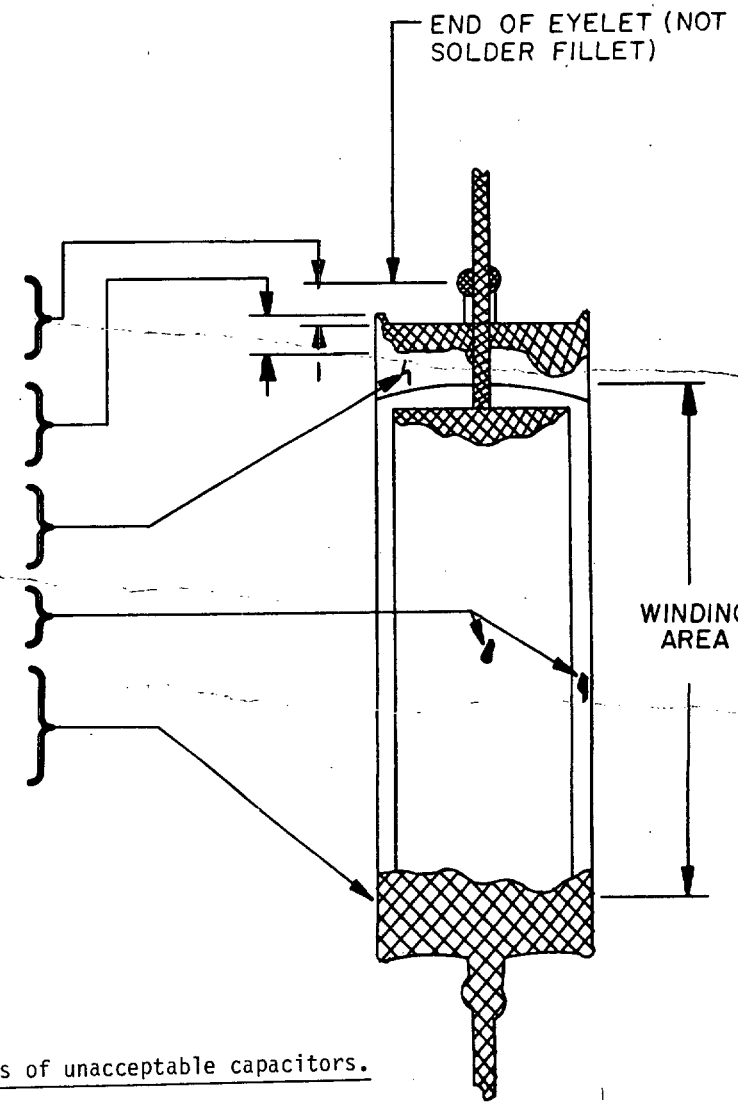


FIGURE 3a. Examples of unacceptable capacitors.

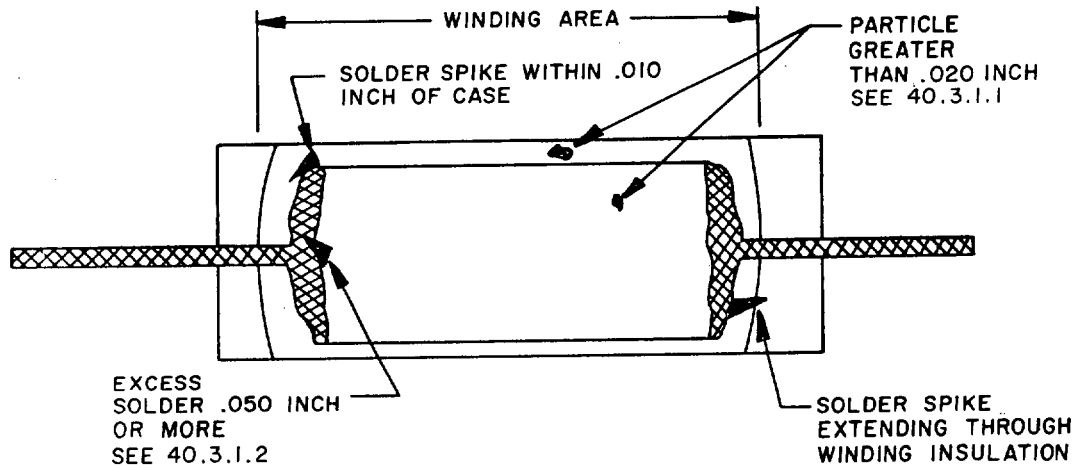
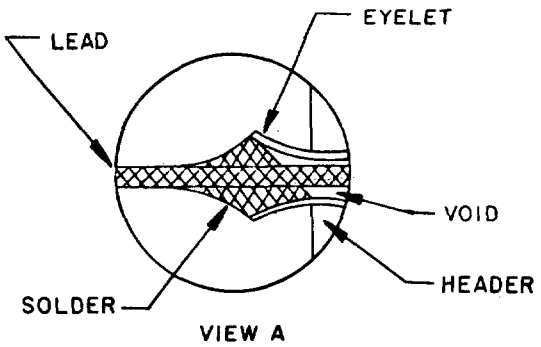


FIGURE 3b. Examples of unacceptable capacitors.

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CROSS SECTION THROUGH
EYELET AND HEADER

Details of End Seal Installation.

The part shall be rejected if any one of the defects listed below is observed.

1. Length from end of eyelet to header less than 50% filled with solder.

NOTE: When excess solder does not permit easy distinction of eyelet end point, the point may be established by radiographic comparison of identical parts on the same film. Two different KV settings may be necessary to accomplish the complete inspection.

2. Solder in header/case joint less than 50% of length of joint.

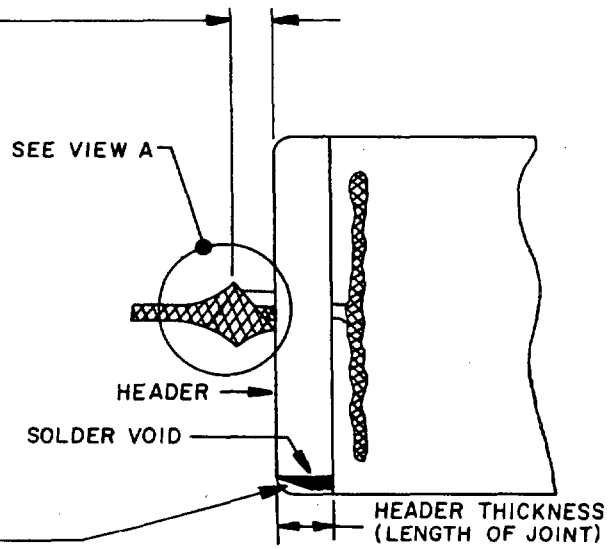


FIGURE 3c. Examples of acceptable and unacceptable capacitors.

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PART EXPOSURE ORIENTATION

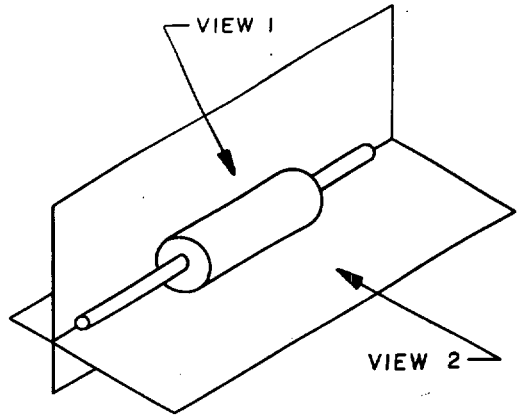


FIGURE 4. Part/Exposure Orientation.

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IQI NO.	TUNGSTEN WIRE DIAMETERS						LEAD PARTICLE DIAMETERS						STEEL SHIM STOCK
	A	B	C	D	E	F	G	H	I	J	K	L	
1	.002	.001	.0005	.0005	.001	.002	.015	.010	.008	.006	.004	.002	None
2	"	"	"	"	"	"	"	"	"	"	"	"	.002
3	"	"	"	"	"	"	"	"	"	"	"	"	.005
4	"	"	"	"	"	"	"	"	"	"	"	"	.007
5	.003	.002	.001	.001	.002	.003	"	"	"	"	"	"	.010
6	"	"	"	"	"	"	"	"	"	"	"	"	.015
7	.005	.003	.002	.002	.003	.005	"	"	"	"	"	"	.025
8	"	"	"	"	"	"	"	"	"	"	"	"	.035

TABLE OF IMAGE QUALITY INDICATORS (IQI)

NOTES:
1. Wires to be tungsten, shim stock to be carbon steel, particles to be lead. Center section to be .125 layers of clear acrylic plastic, bonded with clear plastic cement of low X-ray density. Fasteners may be used within 0.250 or less from each corner, but shall not interfere with end use of the IQI. Bottom surface shall be flush.
2. All dimensions, shown are ±0.005, except wires, and shim stock, which shall be within ±0.001 and lead particles which shall be ±0.0005. Groove details are not critical except that wire must be embedded flush or below surface of plastic and centered at location shown. Particle-hole sizes are not critical, but should not exceed .031 in diameter and depth, and must be centered as shown ±0.005.
3. Additional layers of shim stock may be used as necessary.
4. Identification marking shall be permanent and legible. Location and size of characters is not critical but shall not interfere with or obscure the radiographic image details.

FIGURE 5. Image quality indicator.

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IMAGE QUALITY INDICATOR

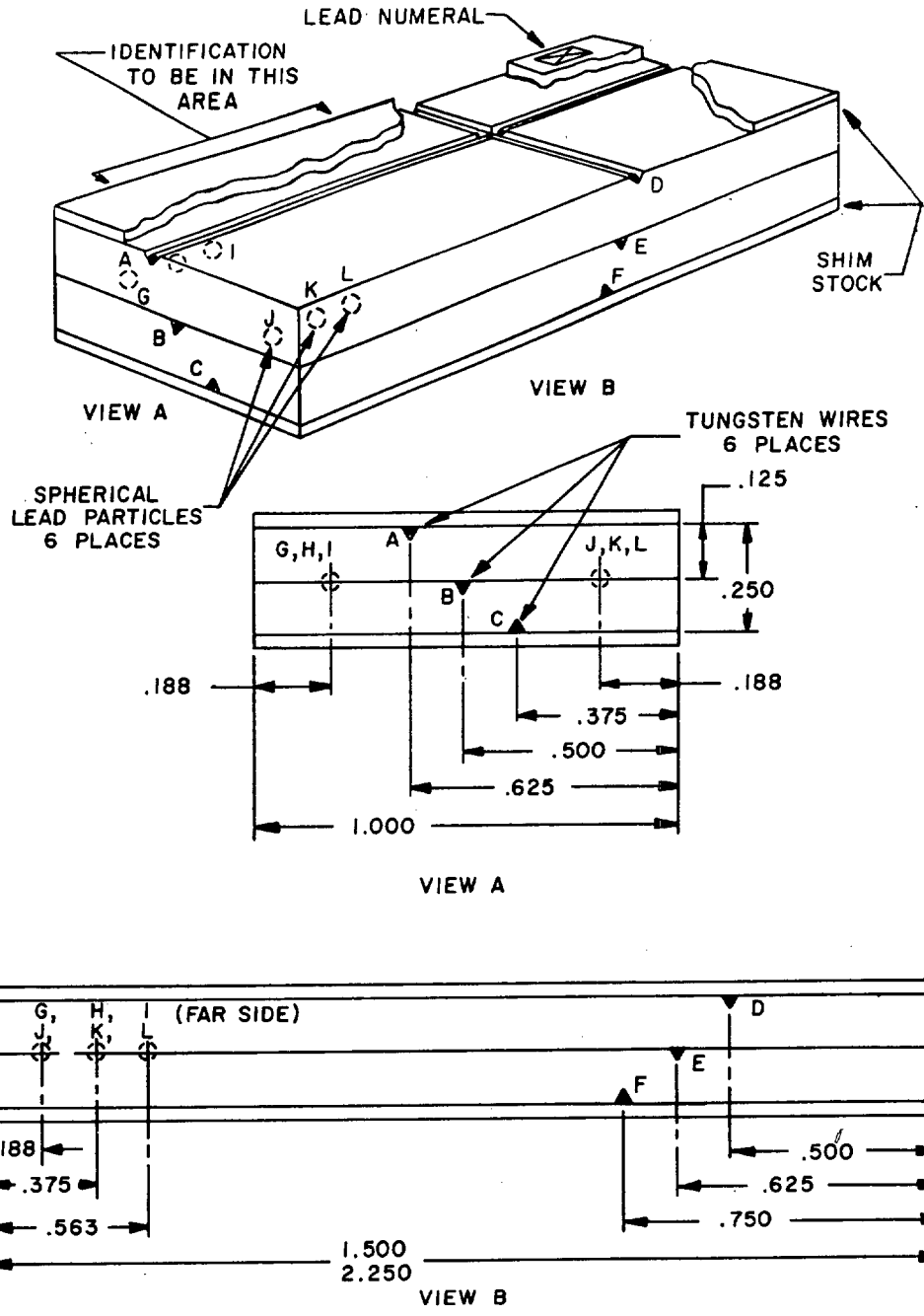


FIGURE 5. Image quality indicator - Continued.

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Image Quality Indicator. Image Quality Indicators (IQI) shall be employed in all radiographic testing and shall be as specified herein (Figure 5). The use of IQI's having only wires is acceptable. However, IQI's having wires and spherical lead articles are preferred. Alternate IQI configurations may be used only with written approval of the preparing activity. The IQI image shall be used to determine radiographic quality. The IQI image shall meet the following requirements:

a. All IQI wires and spherical particles shall be visable on each radiograph.

b. The spacing between wires of the IQI shall not be distorted by more than 10%. The percentage of distortion, as used in this specification, is defined as follows:

$$\text{Percentage Distortion} = \frac{S_1 - S_0}{S_0} = (100)$$

Where S_0 = actual wire spacing.

S_1 = wire spacing as it appears on the X-ray film.

The film density within the image area of the IQI shall match the film density in the readable inspection area of the part image within 10%. When specific IQI configurations are designated in applicable Detail Specification Sheets, the requirement of this paragraph does not apply. Film density shall be determined by means of a densitometer.

FIGURE 5. Image quality indicator - Continued.

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Navy - EC
Air Force - 11
NASA - NA

Review activities:
Army - AR, MI
Air Force - 17, 19, 85
DLA - ES

User activities:
Navy - AS, CG, MC, OS, SH

Agent:
DLA - ES

Preparing activity:
Air Force - 11

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